

JAEHYEOK (EDDIE) LEE

eddie000321@outlook.com | linkedin.com/in/jleedev | github.com/Eddie000321 | Toronto, ON

Summary

- Software Engineering Co-op student focused on applied computer vision, video/signal processing, quality-aware Python pipelines, and SwiftUI data collection.
- Hands-on with Python, OpenCV, SciPy, Swift, SwiftUI, AVFoundation, FFmpeg, Unit Testing.
- Motivated to build non-diagnostic sensing prototypes that surface uncertainty, fail closed on weak inputs, and produce traceable reports and test evidence.

Skills

Databases & Analytics: OpenCV, SciPy, NumPy, pandas, PostgreSQL, Migrations

Languages: Python, Swift, SQL, C#, JavaScript, TypeScript, Java

Frameworks & APIs: SwiftUI, AVFoundation, FastAPI, REST APIs, .NET, Entity Framework Core, Public APIs

Systems & Tools: FFmpeg, Unit Testing, Application Logging, Git, Docker, Azure

Relevant Experience

Support Technician, IT Integrated Services (Co-op)

Jan 2026 - Present

Centennial College | Information Technology Services

Toronto, ON

- **Built an internal operations dashboard** for the team to resolve a **major support workflow bottleneck**, unifying room/lab schedule search, weekly timetable exports, work allocation, and support-state checks across **10.7K schedule events** and **9.6K availability rows**.
- **Created an Excel monitor-upgrade workflow** using Dell service API lookups across **100+ podium asset records**, identifying **29 rooms** needing replacement and tracking room availability as replacement assets were completed.

Selected Projects

Cat Respiration Lab

Python, OpenCV, Swift

- Built a non-diagnostic cat respiration pipeline that normalizes phone video, tracks thorax motion with KLT optical flow, and estimates rate by method consensus.
- Integrated a SwiftUI/AVFoundation iPhone collector with RGB fallback; **25 tests** and a **one-command synthetic demo** cover successful and fail-closed short-input paths.

CareFlow | GitHub

PostgreSQL, .NET, REST API

- Added request logging, health/metrics checks, and **23 tests covering age logic**, deletion guards, an **in-memory HTTP owner/pet workflow**, and migration-model consistency.

Education

Advanced Diploma in Software Engineering Technology Co-op | GPA: 4.30/4.5 Sep 2024 - Expected Dec 2027

Centennial College

Toronto, ON

Key Courses: Testing/QA | C#/.NET | Data Structures & Algorithms | Java | Databases

BSc in Computer Science, Minor in Statistics & Data Science | GPA: 4.4/4.5

Mar 2025 - Expected Dec 2027

Korea National Open University

South Korea

Key Courses: Python | Data Analysis | Database Systems | Algorithms | Java